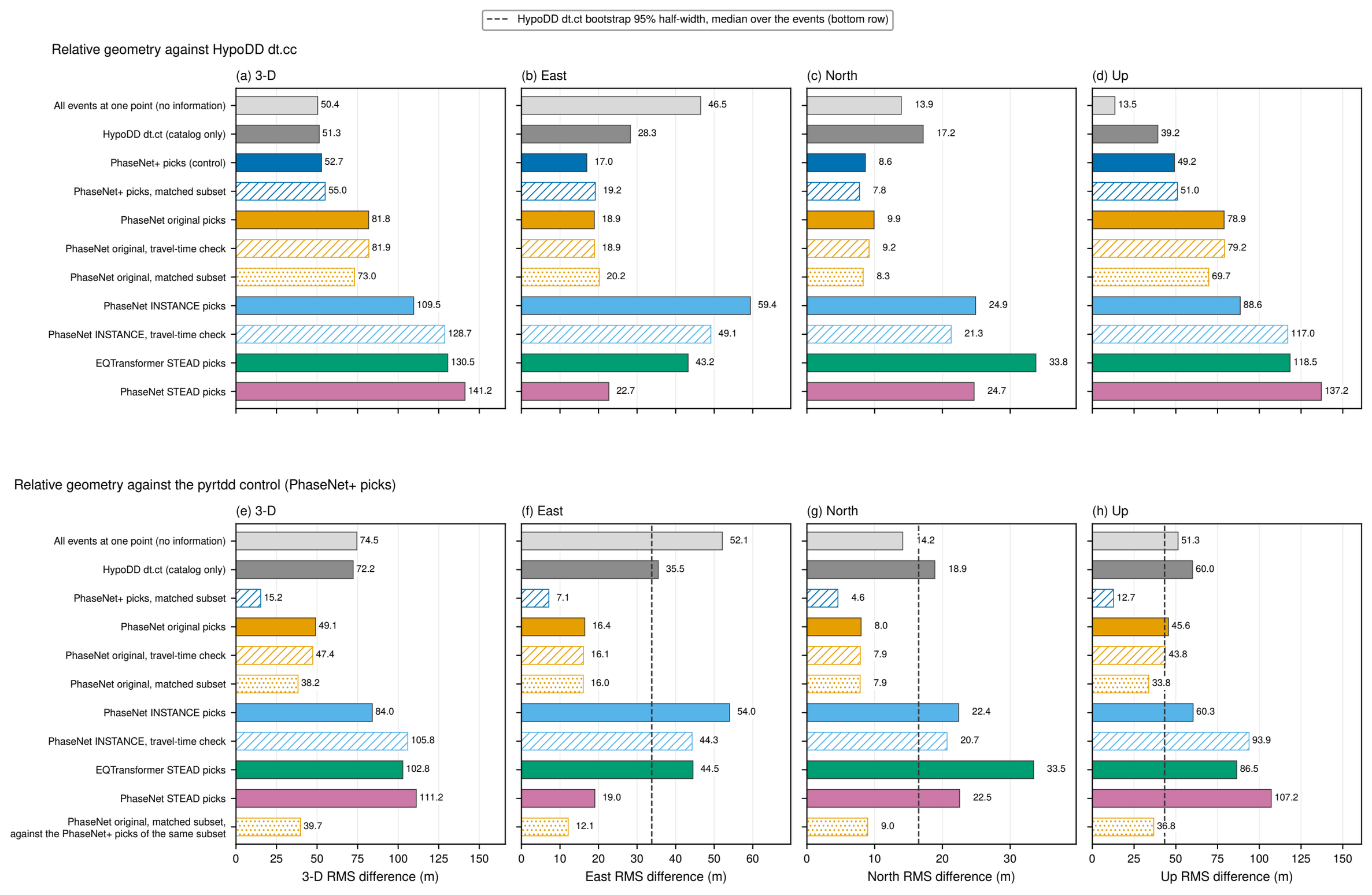

HypoDD dt.ct bootstrap 95% half-width, median over the events (bottom row)
Relative geometry against HypoDD dt.cc
(a) 3-D
(b) East
(c) North
(d) Up
All events at one point (no information)
HypoDD dt.ct (catalog only)
PhaseNet+ picks (control)
PhaseNet+ picks, matched subset
PhaseNet original picks
PhaseNet original, travel-time check
PhaseNet original, matched subset
PhaseNet INSTANCE picks
PhaseNet INSTANCE, travel-time check
EQTransformer STEAD picks
PhaseNet STEAD picks
50.4
51.3
52.7
55.0
81.8
81.9
73.0
109.5
128.7
130.5
141.2
46.5
28.3
17.0
19.2
18.9
18.9
20.2
59.4
49.1
43.2
22.7
13.9
17.2
8.6
7.8
9.9
9.2
8.3
24.9
21.3
33.8
24.7
13.5
39.2
49.2
51.0
78.9
79.2
69.7
88.6
117.0
118.5
137.2
Relative geometry against the pyrtdd control (PhaseNet+ picks)
(e) 3-D
(f) East
(g) North
(h) Up
All events at one point (no information)
HypoDD dt.ct (catalog only)
PhaseNet+ picks, matched subset
PhaseNet original picks
PhaseNet original, travel-time check
PhaseNet original, matched subset
PhaseNet INSTANCE picks
PhaseNet INSTANCE, travel-time check
EQTransformer STEAD picks
PhaseNet STEAD picks
PhaseNet original, matched subset, against the PhaseNet+ picks of the same subset
74.5
72.2
15.2
49.1
47.4
38.2
84.0
105.8
102.8
111.2
39.7
52.1
35.5
7.1
16.4
16.1
16.0
54.0
44.3
44.5
19.0
12.1
14.2
18.9
4.6
8.0
7.9
7.9
22.4
20.7
33.5
22.5
9.0
51.3
60.0
12.7
45.6
43.8
33.8
60.3
93.9
86.5
107.2
36.8
0
25
50
75
100
125
150
3-D RMS difference (m)
0
10
20
30
40
50
60
East RMS difference (m)
0
10
20
30
North RMS difference (m)
0
25
50
75
100
125
150
Up RMS difference (m)